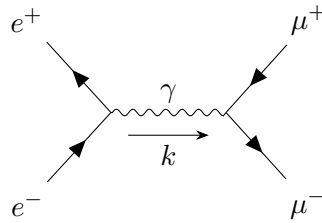


Quantum Field Theory

LECTURE NOTES

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Lecture 30: Wick's Theorem

In the previous section we looked into an introduction to interacting fields and found general expressions for the correlation functions

$$\langle \Omega | \mathcal{T} \{ \phi(x_1) \phi(x_2) \dots \phi(x_n) \} | \Omega \rangle$$

where we need to find the time-ordered product of free fields. These are efficiently given by a method of computation. As a gross approximation, Wick's theorem tells us how to go from time ordered products to normal ordered products.

Recall from sec ?? that normal ordered products meant that all the creation operators are sent to the left. This is particularly useful, since if creation operators are on the left (and annihilation are on the right), then these will kill the vacuum from both sides when expectation is taken, which might simplify our calculations somewhat.

We will use the notation $\phi(x_i) \equiv \phi_i$ and for the Feynman propagator, $D_F(x_i - x_j) \equiv \Delta_{ij}$

1.1 Few calculations

For $n = 1$, $\mathcal{T}\{\phi_1\} = \phi_1$ since there is only one field. For $n = 2$,

$$\mathcal{T}\{\phi_1, \phi_2\} = \Theta(x_1^0 - x_2^0) \phi_1 + \Theta(x_2^0 - x_1^0) \phi_2$$

which we already know, however, let us see this a bit differently. For that, decompose the Heisenberg field $\phi(x)$ into $\phi^+(x) + \phi^-(x)$ where

$$\begin{aligned} \phi^+(x) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \hat{a}_{\mathbf{p}} e^{-ip \cdot x} \\ \phi^-(x) &:= \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \hat{a}_{\mathbf{p}}^\dagger e^{ip \cdot x} \end{aligned}$$

Note the unfortunate notation that $+$ comes with annihilation and $-$ comes with creation operators. Normal ordering is there if minuses (if any) are to the left. Now suppose that we assume the time-order $x_1^0 > x_2^0$ which gives

$$\begin{aligned} \mathcal{T}\{\phi_1, \phi_2\} &= \phi_1 \phi_2 = (\phi_1^+ + \phi_1^-)(\phi_2^+ + \phi_2^-) \\ &= \phi_1^+ \phi_2^+ + \phi_1^+ \phi_2^- + \phi_1^- \phi_2^+ + \phi_1^- \phi_2^- \\ &= \phi_1^+ \phi_2^+ + \phi_2^- \phi_1^+ + \phi_1^- \phi_2^+ + \phi_1^- \phi_2^- + [\phi_1^+, \phi_2^-] \end{aligned}$$

The blue term was not normal ordered and hence in the last line, we made that normal ordered by introducing the commutator. The commutator gives us $[\phi_1^+, \phi_2^-] = D(x - y) \equiv D_{12}$ where $D(x - y)$ is defined as in sec ??.

As a shorthand, we denote the normal-ordering by $\mathcal{N}\{\phi_1, \phi_2\} \equiv : \phi_1 \phi_2 :$, hence for $x_1^0 > x_2^0$

$$\mathcal{T}\{\phi_1, \phi_2\} = : \phi_1 \phi_2 : + D_{12}$$

If we calculate for the other case, $x_1^0 < x_2^0$ then 1 and 2 are simply exchanged,

$$\mathcal{T}\{\phi_1, \phi_2\} = : \phi_1 \phi_2 : + D_{21}$$

Thus in general we can write the time-ordered product in the following form

$$\begin{aligned} \mathcal{T}\{\phi_1, \phi_2\} &= \Theta(x_2^0 - x_1^0) [: \phi_1 \phi_2 : + D_{21}] + \Theta(x_1^0 - x_2^0) [: \phi_1 \phi_2 : + D_{12}] \\ &= : \phi_1 \phi_2 : + \Theta(x_2^0 - x_1^0) D_{21} + \Theta(x_1^0 - x_2^0) D_{12} \\ &= : \phi_1 \phi_2 : + \Delta_{12} \end{aligned}$$

We should have been more careful with the above expression. The LHS is an operator, however the Feynman propagator Δ_{ij} is just a complex number, hence we should have technically written $\Delta_{12}\mathbb{1}$. We define a *contraction* as

$$\overline{\phi_1\phi_2} := \Delta_{12}\mathbb{1}$$

Then the time-ordered product of two-fields can be written as

$$\mathcal{T}\{\phi_1, \phi_2\} = : \phi_1\phi_2 : + \overline{\phi_1\phi_2}$$

1.2 The main theorem

Theorem 1 (Wick's Theorem):

The time-ordered pairs of fields is given by

$$\mathcal{T}\{\phi_1, \phi_2 \dots \phi_n\} = \sum_{m=0}^{\lfloor n/2 \rfloor} \sum_{\text{all } m \text{ pairings}} : \underbrace{\phi_1\phi_2 \dots \phi_n}_{m \text{ contractions}} :$$

What the theorem says is that the time-ordered product of n fields is the equal to the sum of operators formed by all possible contractions between the fields.

We now try to prove the theorem but before that let us see an example for $n = 4$. Note that any contraction will remove 2 fields and hence the maximum number of contractions possible is $\lfloor n/2 \rfloor$. If n is odd, then one field is always uncontracted.

Applying this to four fields, we get

$$\begin{aligned} \mathcal{T}\{\phi_1, \phi_2, \phi_3, \phi_4\} = & : \phi_1\phi_2\phi_3\phi_4 : + \overline{\phi_1\phi_2}\phi_3\phi_4 + \overline{\phi_1\phi_3}\phi_2\phi_4 + \overline{\phi_1\phi_4}\phi_2\phi_3 + \overline{\phi_2\phi_3}\phi_1\phi_4 \\ & + \overline{\phi_2\phi_4}\phi_1\phi_3 + \overline{\phi_3\phi_4}\phi_1\phi_2 + \overline{\phi_1\phi_2}\phi_3\phi_4 + \overline{\phi_1\phi_3}\phi_2\phi_4 + \overline{\phi_1\phi_4}\phi_2\phi_3 \\ & + \overline{\phi_2\phi_3}\phi_1\phi_4 + \overline{\phi_2\phi_4}\phi_1\phi_3 + \overline{\phi_3\phi_4}\phi_1\phi_2 \end{aligned}$$

where the contraction between two fields not adjacent to each other, like say $\overline{\phi_1\phi_3}\phi_2\phi_4$, means that the Feynman propagator is between those two fields and the other fields are normal-ordered, that is,

$$(\Delta_{13})(: \phi_2\phi_4 :)$$

Proof. We have already proved it for $n = 1$ and $n = 2$. Let fields $\phi_1, \dots, \phi_n$ be given and WLOG we can relabel the fields such that these become time-ordered, that is, $x_1^0 \geq x_2^0 \dots \geq x_n^0$ and we claim that Wick's theorem is true for $(n - 1)$ fields, $\phi_2, \phi_3 \dots \phi_n$, that is

$$\mathcal{T}\{\phi_2 \dots \phi_n\} = \sum_{m=0}^{\lfloor (n-1)/2 \rfloor} \sum_{\text{all } m \text{ pairings}} : \underbrace{\phi_2 \dots \phi_n}_{m \text{ contractions}} : \equiv \mathcal{W}(\phi_2 \dots \phi_n)$$

We have to now show that it is true for n fields. Multiply both sides of the above equation by ϕ_1 . Since x_1^0 is greater than all other terms, the LHS becomes $\mathcal{T}\{\phi_1\phi_2 \dots \phi_n\}$. Now the RHS becomes

$$\phi_1 \mathcal{W}(\phi_2 \dots \phi_n) = (\phi_1^+ + \phi_1^-) \mathcal{W}(\phi_2 \dots \phi_n)$$

The second term is not problematic, since all the terms in $\mathcal{W}$ are already normal ordered, so ϕ_1^- can safely move in to the left, without disturbing the normal-ordering. For ϕ_1^+ we need to send it to the extreme right and thus, we have to invoke the commutator.

$$\phi_1 \mathcal{W}(\phi_2 \dots \phi_n) = \phi_1^- \mathcal{W}(\phi_2 \dots \phi_n) + \mathcal{W}(\phi_2 \dots \phi_n) \phi_1^+ + [\phi_1^+, \mathcal{W}(\phi_2 \dots \phi_n)]$$

The first two terms together contain all possible contractions without ϕ_1 and multiplied with ϕ_1 , so it has all possible normal-ordered cases where ϕ_1 is not contracted. Now consider the commutator part

(the third term). Note that $\mathcal{W}(\phi_2 \dots \phi_n)$ contains all contractions which do not contain ϕ_1 . So it has a general form of some contraction times a normal-ordered product of fields,

$$(\text{contraction}) [\phi_1^+, \phi_{i_1} \phi_{i_2} \dots \phi_{i_k}]$$

Now take any field ϕ_{i_l} and decompose into $\phi_{i_l}^+$ and $\phi_{i_l}^-$. Commutator of ϕ_1^+ with the first term is simply zero while the commutator with the second term produces a contraction D_{1i_l} . In this way, the commutator part produces all contractions containing ϕ_1 . Thus, the entire term produces all contractions with fields $\phi_1 \dots \phi_n$ and we showed that it is true for n fields, starting with the assumption that it is true for $n - 1$ fields. By induction this is true for all $n \geq 1$ and hence proved.

Note that when we take the vacuum expectation of the time-ordered product, all the terms with a dangling normal ordering is killed, since the creation and annihilation operators act on the vacuum from the right direction so as to kill it. Only those term survive where all the fields have been contracted. For $n = 4$ only the blue terms survive which can be diagrammatically represented as

$$\langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi_3 \phi_4 \} | \Omega \rangle = \begin{array}{c} 1 \bullet \text{---} \bullet 2 \\ 3 \bullet \text{---} \bullet 4 \end{array} + \begin{array}{c} 1 \bullet \\ | \\ 3 \bullet \end{array} \begin{array}{c} 2 \bullet \\ | \\ 4 \bullet \end{array} + \begin{array}{c} 1 \bullet \text{---} \bullet 2 \\ \diagdown \quad \diagup \\ 3 \bullet \text{---} \bullet 4 \end{array}$$

The diagrams represents the process of particle creation, propagation, and annihilation which takes place in spacetime. Each line denotes a propagator and the nodes are the starting or ending point of each process.

Lecture 31: Feynman diagrams

In the previous case, we discussed about Wick's theorem and time-ordered product of fields can be simply written as a sum of contractions of the fields. We will now consider the case of ϕ^4 interacting theory and try to find the n -point correlation functions of the interacting theory.

Recall from Eq. ?? that the correlation functions are basically the vacuum expectation of the time-ordered field products and the time-ordered product is known from Wick's theorem. Thus we already have a recipe to calculate this and we will use a perturbative approach for this.

We had considered the interaction of the form ϕ^4 and from this $\mathcal{H}_I = \frac{\lambda}{4!} \phi^4$ where everything is in the interaction picture. Let us consider the **numerator** for the two-point correlation function first which is of the form

$$\langle \Omega | \mathcal{T} \left\{ \phi_1 \phi_2 \exp \left(\frac{-i\lambda}{4!} \int d^4 z \phi^4(z) \right) \right\} | \Omega \rangle$$

We first Taylor expand the exponential

$$\exp \left(\frac{-i\lambda}{4!} \int d^4 z \phi^4(z) \right) = \mathbb{1} + \left(\frac{-i\lambda}{4!} \right) \int d^4 z \phi^4(z) + \left(\frac{-i\lambda}{4!} \right)^2 \int d^4 z d^4 w \phi^4(z) \phi^4(w) + \dots$$

and then the numerator takes on a ghastly form

$$\langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \} | \Omega \rangle + \left(\frac{-i\lambda}{4!} \right) \int d^4 z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \} | \Omega \rangle + \frac{1}{2!} \left(\frac{-i\lambda}{4!} \right)^2 \int d^4 z d^4 w \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \phi^4(w) \} | \Omega \rangle + \dots$$

It is now possible to use Wick's theorem to characterise all these integrals, carefully considering the cumbersome contractions.

▷ Term of $\mathcal{O}(1)$:

The term is $\langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \} | \Omega \rangle$, which is simply Δ_{12} and is denoted diagrammatically as

$$\begin{array}{c} \bullet \text{---} \bullet \\ x \quad y \end{array}$$

▷ Term of $\mathcal{O}(\lambda)$:

We have to work a little bit here. The term is

$$\int d^4z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \} | \Omega \rangle \equiv \int d^4z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi(z) \phi(z) \phi(z) \phi(z) \} | \Omega \rangle$$

We have to find all type of contractions amidst these six fields if we use the Wick's theorem. Note that any partial contractions will not matter, since we are taking vacuum expectations and uncontracted normal ordered products will kill it. Thus we need to consider only fully contracted fields. How many such configurations are there?

It turns out that we need to do some combinatorics for this. Indeed, all *good* math and physics ends in either linear algebra or combinatorics¹.

So if we take one field, then it can be contracted with the remaining 5, then these two fields are gone, remaining four are there. If we choose one, we can contract with remaining 3, and then the last two automatically gets contracted. So total $5 \times 3 = 15$ such fully contracted field configurations.

Due to symmetry conditions, we only have two distinct types of contractions which are,

$$\overbrace{\phi_1 \phi_2} \quad \overbrace{\phi(z) \phi(z)} \quad \overbrace{\phi(z) \phi(z)} \quad \overbrace{\phi_1 \phi_2 \phi(z) \phi(z) \phi(z) \phi(z)}$$

For the first case, we have $x - y$ contracting with each other and for the second case, we have $x - z$ contracting. These can be diagrammatically represented as:

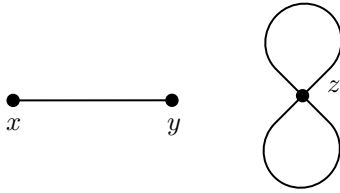


Figure 1: Type I contraction

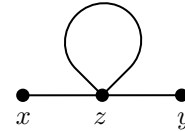


Figure 2: Type II contraction

There are three of the above Type I terms, since if x contracts with y then the first z can contract with three other guys.

Let us consider the second type of contraction where x contracts with z . There are 12 such terms since x can go with z in 4 ways and for each xz pair there are three ways y can go with z . Then the term of order $\mathcal{O}(\lambda)$ becomes

$$3 \times \left(\frac{-i\lambda}{4!} \right) \int d^4z \Delta^2(x-y) \Delta^2(z-z) + 12 \times \left(\frac{-i\lambda}{4!} \right) \int d^4z \Delta(x-z) \Delta(y-z) \Delta(z-z)$$

(For now just take $\Delta(z-z)$ to be some ‘number’ (even though it diverges) and let us not worry about this!)

Some observations

- ▷ The outermost nodes in the diagram (Eg. x and y) are the *external vertices*. These vertices are not integrated over.
- ▷ The inner nodes (Eg. z) are called the *internal vertices* and for each internal vertex, we have to associate an integral to them.

We had found the numerator upto order λ and now let us find the **denominator**.

$$1 + \left(\frac{-i\lambda}{4!} \right) \int d^4z \langle \Omega | \mathcal{T} \{ \phi^4(z) \} | \Omega \rangle + \frac{1}{2!} \left(\frac{-i\lambda}{4!} \right)^2 \int d^4z d^4w \langle \Omega | \mathcal{T} \{ \phi^4(z) \phi^4(w) \} | \Omega \rangle + \dots$$

¹which implies that if you don't find yourself doing these, you are not doing good physics

Note that the denominator does not have any free fields ϕ_1 and ϕ_2 and thus, the $\mathcal{O}(1)$ term will be just 1 while the term of $\mathcal{O}(\lambda)$ will consist of only the contractions with z (3 terms). Thus symbolically (since the integrals will be just some number) we can write the two point correlator function as:

$$\begin{aligned}
 & \frac{\text{diagram}(x,y) + 3 \left(\text{diagram}(x,y,z) \right) + 12 \left(\text{diagram}(x,z,y) \right) + \mathcal{O}(\lambda^2)}{1 + 3 \text{diagram}(z) + \mathcal{O}(\lambda^2)} \\
 &= \left[\text{diagram}(x,y) + 3 \left(\text{diagram}(x,y,z) \right) + 12 \left(\text{diagram}(x,z,y) \right) + \mathcal{O}(\lambda^2) \right] \left[1 - 3 \text{diagram}(z) + \mathcal{O}(\lambda^2) \right] \\
 &= \text{diagram}(x,y) - 3 \left(\text{diagram}(x,y,z) \right) + 3 \left(\text{diagram}(x,y,z) \right) + 12 \left(\text{diagram}(x,z,y) \right) + \mathcal{O}(\lambda^2) \\
 &= \left[\text{diagram}(x,y) + 12 \left(\text{diagram}(x,z,y) \right) + \mathcal{O}(\lambda^2) \right]
 \end{aligned}$$

Note that, a cancellation was involved in the reduction such that the diagram with two disconnected components got cancelled. Even though this seems like a miracle, in fact, it's not!

Turns out, only those diagrams contribute where the components are connected to the external points. The contributions from all the disconnected subgraphs cancel. Before showing this, let us write a summary of what we need to do for systematic calculations:

- For each propagator $\text{diagram}(x,y)$ we associate the Feynman propagator $\Delta(x-y)$.
- For each internal vertex $\text{diagram}(z)$ we associate the term $(-i\lambda) \int d^4z$ (that is, we integrate over any internal vertices).
- For each external vertex, we associate the value 1.
- Finally we divide by the *symmetry factor*, which is like an *enigmatic term* but basically means the number of ways we can rearrange the propagators or vertices, without changing the diagram (Peskin says that we *almost surely* never need to calculate any diagram with symmetry factor greater than 2).

The above set of rules are called the *Feynman rules* in position-space (since we are considering the fields in the real space) and these are specific to the ϕ^4 -theory. For other type of interactions, few changes are accordingly made.

Theorem 2 (Linked-Cluster Theorem):

$$\langle \tilde{\Omega} | \mathcal{T} \{ \phi_1 \dots \phi_n \} | \tilde{\Omega} \rangle = \left(\begin{array}{l} \text{sum of all connected diagrams} \\ \text{subjected to Feynman rules} \end{array} \right)$$

Proof. A typical diagram consists of *connected* and *disconnected* diagrams, where connected diagrams are the ones where everything is connected to some external vertices. Disconnected diagrams are such that the diagrams consist of different components where at least one component is not connected to any external vertex. The components not connected to any external vertices are called the *vacuum bubbles*.

Let us consider the set of all vacuum bubbles represented by $\mathcal{V}$, then we have

$$\mathcal{V} = \left\{ \text{bubble} , \text{chain of bubbles} , \text{bubble with internal lines} , \text{etc.} \right\}$$

Suppose a diagram V (where V actually denotes the value of the integral represented by the diagram) has n_i pieces of $v_i \in \mathcal{V}$, in addition to the connected components (containing external vertices) whose value is, let say, c . Then the contribution of the diagram becomes:

$$c \cdot \prod_i \frac{1}{n_i!} (v_i)^{n_i}$$

where the $n_i!$ is taken since there are n_i copies of v_i , but these copies are indistinguishable. Then the numerator of the correlation function is obtained by summing over all the different connected components c

$$\text{numerator} = \sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} \sum_{\{n_i\}} c \cdot \prod_i \frac{1}{n_i!} (v_i)^{n_i} = \sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} c \cdot \prod_i \sum_{\{n_i\}} \frac{1}{n_i!} (v_i)^{n_i} = \left(\sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} c \right) \cdot \left(\exp \left(\sum_i v_i \right) \right)$$

where we have used the expansion for e^x and we finally obtain that the numerator is just the product of the connected contribution and the vacuum bubble contribution. The denominator has no connected contributions since the external fields do not occur in the denominator. Hence,

$$\text{denominator} = \exp \left(\sum_i v_i \right)$$

The exponential part thus cancels when we divide the numerator and the denominator, leaving only the contribution from the connected components.

NOTE: Higher order (> 2 point) correlation functions may have apparent disconnected pieces in the diagram, like say, the diagrams for the four point correlation function in [1.2](#), where two of the diagrams are not connected. However, each disconnected piece is itself connected to some or the other external vertices. When we talk about disconnected pieces (or vacuum bubbles), we talk about diagrams which contain a piece where that piece has no connection with any of the external vertex.